GM5aWv1

Directions: List the factors to find the GCF. Then use it to reduce to lowest terms.

24	
56	

GCF = _____ $\frac{24}{56}$ = _____

Directions: Solve.

Tanya walked $\frac{3}{5}$ of a mile today and $\frac{6}{8}$ of a mile yesterday. How far did she walk in all?

Level 5a

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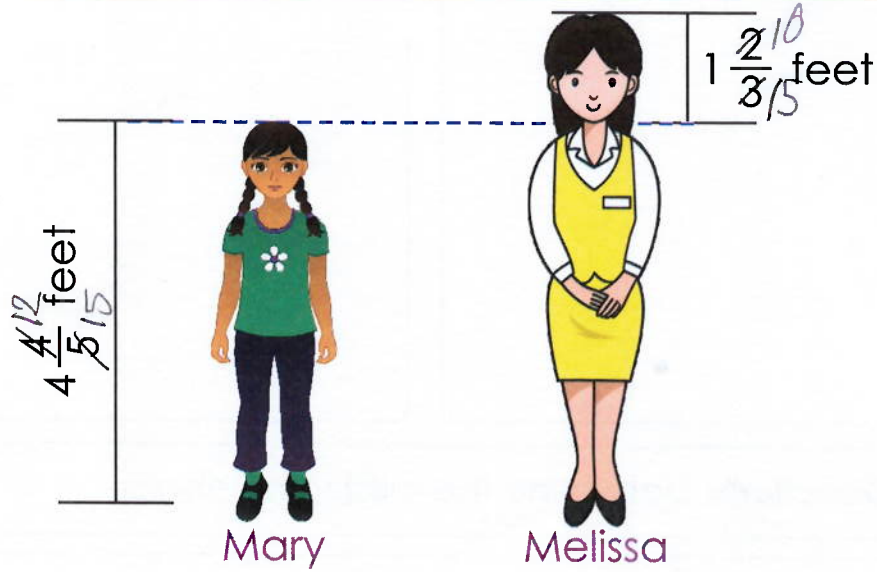
Name Ojas

Date / /

Time ____ : ____ TO ____ : ____

Week 23 W1

Fractions: All Operations **Ratios and Proportions**



$4\frac{12}{15} + 1\frac{10}{15} = 5\frac{22}{15} = 6\frac{7}{15}$ ft.

How tall is Melissa?

a. $6\frac{7}{15}$ feet b. $5\frac{7}{15}$ feet

c. $6\frac{3}{15}$ feet d. $6\frac{7}{18}$ feet

Teacher's Note

All answers in this week's assignment must be written as proper fractions in lowest terms.

Directions: Solve. Express in simplest form.

1. $3 \times \frac{1}{4} = \frac{3}{4}$

2. $2\frac{4}{8} \times 1\frac{1}{3} = 3\frac{1}{3}$

$\frac{20}{8} \times \frac{4}{3} = \frac{80}{24} =$

$3\frac{8}{24} = 3\frac{1}{3}$

Directions: Determine the missing numbers.

(1) $\boxed{2} \div \frac{1}{4} = 8$

(2) $4 \div \frac{\boxed{1}}{\boxed{5}} = 20$

GM5aW1.2

Directions: Find the LCM for each set of numbers.

1. 3, 8, 4

2. 4, 3, 6

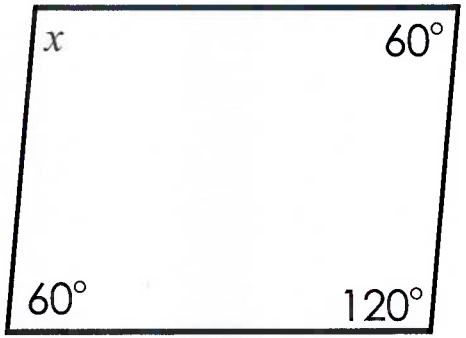
3. 5, 7, 10

4. What is the least common multiple of 12, 6 and 10?

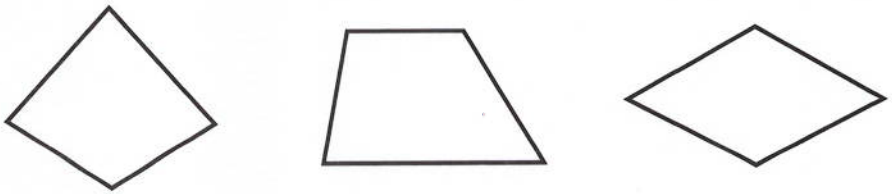
a. 120 b. 30 c. 60 d. 2

GM5aW7.7

Directions: The sum of the interior angles of a rhombus is 360°. Find the missing angle.



Directions: Circle all the trapezoids. Place an **X** on all the rhombuses.



Iman spent three days preparing for her exam. The chart shows the amount of time in hours that she spent.

Day	Hours
Mon.	$1\frac{\cancel{3}15}{\cancel{4}20}$
Tues.	$1\frac{\cancel{4}16}{\cancel{5}20}$
Wed.	$\frac{\cancel{2}4}{\cancel{10}20}$

a. How long did Iman spend studying on all three days?

$1\frac{15}{20} + 1\frac{16}{20} + \frac{4}{20} = 2\frac{35}{20} =$

$3\frac{15}{20} = 3\frac{3}{4}$

Directions: Find the average of the numbers below.

$\frac{\cancel{1}2}{\cancel{4}8}, \frac{1}{8}, \frac{\cancel{3}6}{\cancel{4}8}, \frac{\cancel{4}2}{\cancel{16}8}$

$\frac{2}{8} + \frac{1}{8} + \frac{6}{8} + \frac{2}{8} = \frac{11}{8} = 1\frac{3}{8}$

$\frac{11}{8} \div \frac{4}{1} = \frac{11}{8} \times \frac{1}{4} = \frac{11}{32}$

$\frac{11}{32}$

Which is the least common denominator when simplifying $\frac{2}{3} + 2\frac{1}{4} + \frac{3}{6}$?

2 | 3, 4, 6
3 | 3, 2, 3
2 | 1, 2, 1
1, 1, 1

2 X 3 X 2
12

a. 6 b. 24 c. 10 d. 12

Directions: Evaluate when $x = 5$ and $y = 3$.

1

$y \div \frac{2}{4} =$

$\frac{3}{1} \times \frac{4}{2} = \frac{12}{2} = 6$

2

$5 \div \frac{1}{x} =$

$5 \times \frac{5}{1} = \frac{25}{1} = 25$

GM5aW1.4

Directions: Convert to an improper fraction. Then use the butterfly model to solve.

$9\frac{4}{5} - 1\frac{6}{10} =$

My dog is $7\frac{3}{5}$ years old. My cat is $2\frac{1}{2}$ years old. How much older is my dog than my cat?

GM5aW7.5

Directions: Write the following fractions in the form of ratios using symbols.

(1) $\frac{2}{16}$ = _____

(2) $\frac{8}{9}$ = _____

(3) $\frac{3}{10}$ = _____

(4) $\frac{6}{11}$ = _____

(5) $\frac{4}{20}$ = _____

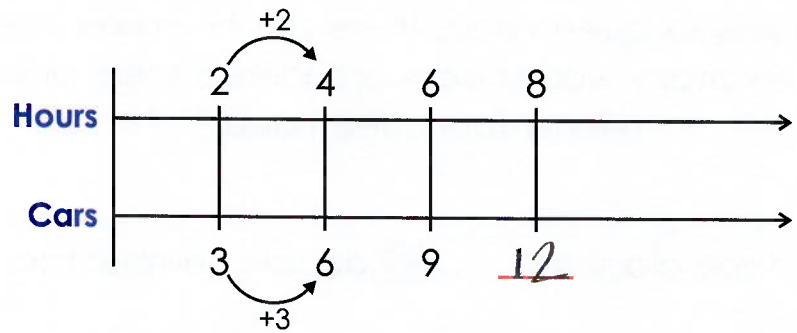
(6) $\frac{1}{2}$ = _____

Directions: Write an example and draw a picture to match the ratio 8:10.

Example	Illustration

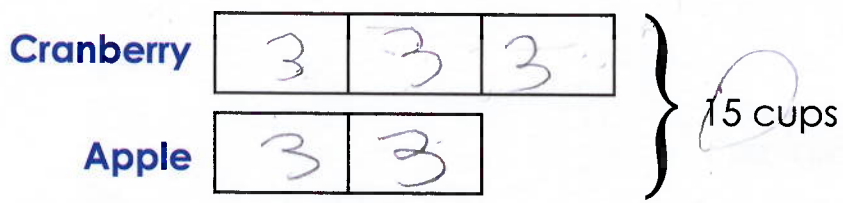
Double number lines are best used when comparing two items with **different** units.

Nina spent 2 hours washing 3 cars. Assuming she works at a constant rate for 8 hours, how many cars will she wash?



Tape diagrams are the best way to model ratios when the **same** type of units are used.

To make fruit punch you need 3 cups of cranberry juice for 2 cups of apple juice.



What does each box represent if you need 15 total cups?
15 ÷ 5 = 3; therefore each box now represents 3 cups.

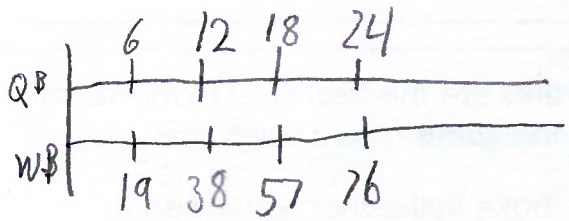
So we would need 9 cups of cranberry juice and 6 cups of apple juice.

If student needs more practice with ratios and proportions, review 5a-P.

Directions: Choose whether a double number or tape diagram would be best used to answer the question. Then explain your choice and solve.

For every six queen bees there are 19 worker bees. How many worker bees are there if there are twenty-four queen bees?

☐ tape diagram ☒ double number line



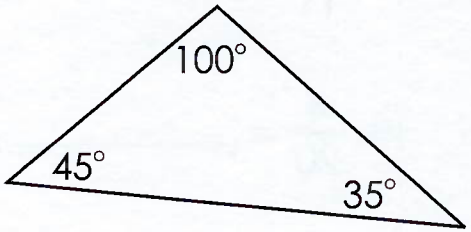
Explain

I am comparing with two different units.

Answer: 76

GM5aW1.6

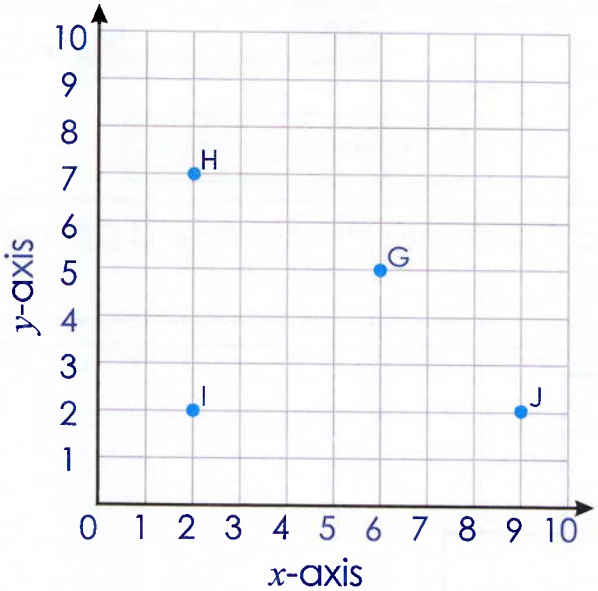
Directions: Identify the triangle from the angle measurements.



Directions: Complete the chart.

Sides of Triangles	Triangle Type
4cm, 4cm, 4cm	
11in, 9in, 7in	
5m, 8m, 8m	

GM5aW7.3



Graph the following points on the coordinate plane.

(1) A (0, 8) (3) C (4, 3) (5) E (5, 0)

(2) B (2, 1) (4) D (3, 7) (6) F (6, 4)

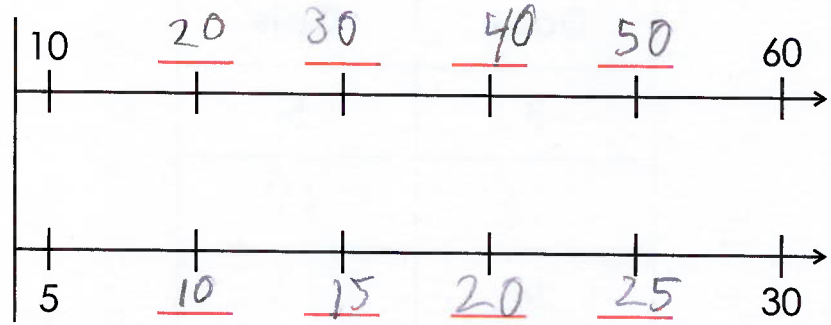
Identify the following points on the coordinate plane.

(1) G: _____ (3) I: _____

(2) H: _____ (4) J: _____

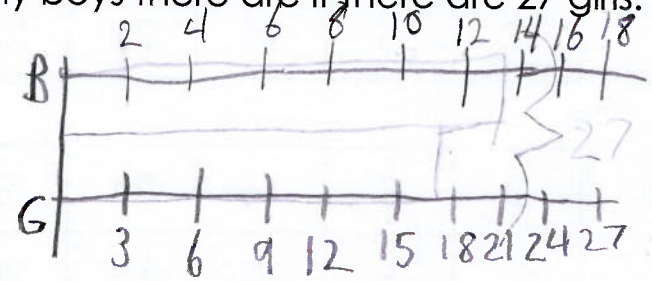
GM5aW7.2

Directions: Complete the double number line.



Directions: Use any method to solve.

The ratio of boys to girls on a school field trip is 2:3. How many boys there are if there are 27 girls.



18 boys

GM5aW1.7

Which table correctly completes the ratio table?

Dogs	Cats
8	5
16	10
24	15
32	20
40	25

(a)

D	8	16	24	32	40
C	5	15	20	25	30

(b)

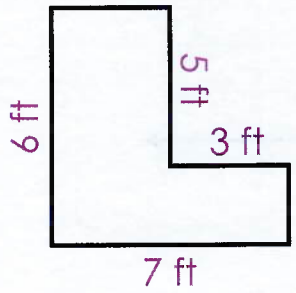
D	8	16	24	32	40
C	5	10	15	20	25

(c)

D	8	16	24	32	40
C	5	15	20	25	30

Level Review

Directions: Calculate the area and perimeter. Show your work.



area: _____ perimeter: _____

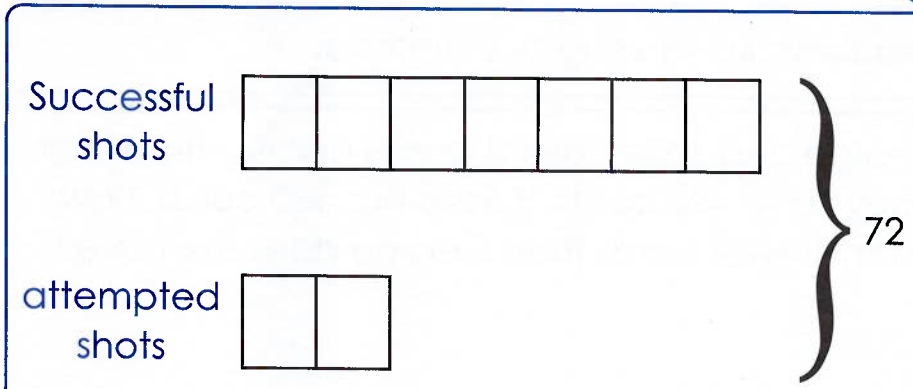
The perimeter of a rectangle is 24 feet. What is the area of the rectangle if the length of one side is 5 feet?

a. 7 feet squared

b. 16 feet squared

c. 35 feet squared

d. 12 feet squared



Liam is a superb basketball player. Above shows the number of successful shots and attempted shots in the last two games. How many shots did he make?

a. 9 shots

b. 16 shots

c. 56 shots

d. 21 shots

Level 5a

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Name

Date / /

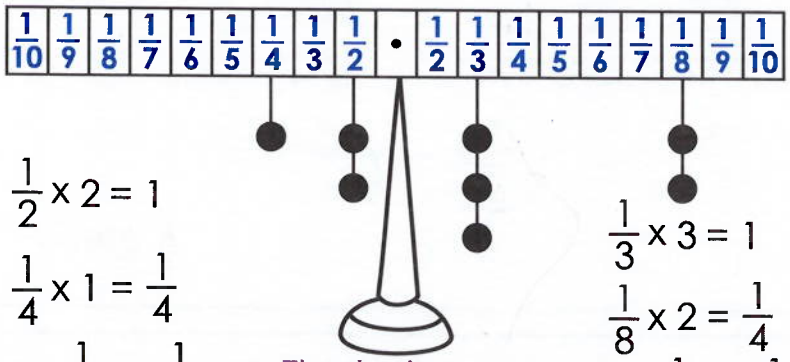
Time ____ : ____ TO ____ : ____

Week 23 W2

Fractions: All Operations | **Ratios and Proportions**

Directions: Level the balance scale.

Ex:



$\frac{1}{2} \times 2 = 1$

$\frac{1}{4} \times 1 = \frac{1}{4}$

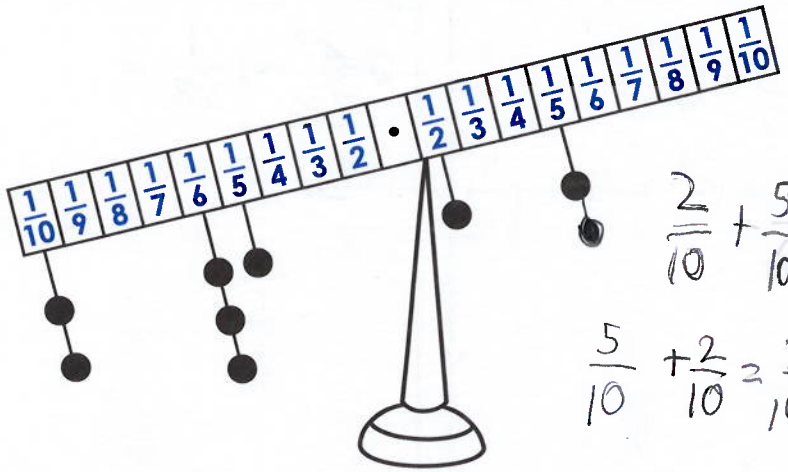
$1 + \frac{1}{4} = 1\frac{1}{4}$

The balance is equal.

$\frac{1}{3} \times 3 = 1$

$\frac{1}{8} \times 2 = \frac{1}{4}$

$1 + \frac{1}{4} = 1\frac{1}{4}$



$\frac{2}{10} + \frac{5}{10} + \frac{2}{10} = \frac{9}{10}$

$\frac{5}{10} + \frac{2}{10} = \frac{7}{10}$ $\frac{9}{10} + \frac{7}{10} = \frac{16}{10}$

$= 1\frac{6}{10} = 1\frac{3}{5}$

What fraction would you hang the ● from to balance the scale?

Joyce made $4\frac{4}{5}$ pounds of trail mix. If she puts $\frac{4}{5}$ of a pound into each bowl, how many bowls will she need to put out all of the trail mix?

$4\frac{4}{5} \div \frac{4}{5} =$

$\frac{24}{5} \div \frac{4}{5} =$

$\frac{24}{5} \times \frac{5}{4} = \frac{120}{20} = 6$

6 bowls

Directions: Solve using any method.

1

$4\frac{\cancel{1}5}{\cancel{3}15} + 6\frac{\cancel{2}6}{\cancel{5}15} =$

$\frac{65}{15} + \frac{96}{15} = \frac{161}{15} = 1\frac{11}{15}$

$1\frac{11}{15}$

2

$3\frac{\cancel{2}8}{\cancel{7}28} - 1\frac{1}{28} =$

$\frac{92}{28} - \frac{29}{28} = \frac{63}{28} = 2\frac{7}{28}$

$2\frac{1}{4}$

$2\frac{1}{4}$

Directions: Solve using any method.

George and Josie collect game cards. Altogether they have 490 cards. If Josie has 180 cards, how many fewer cards than George does she have?

Write a ratio to describe George and Josie's cards in simplest form.

Lakisha earns $180 in 12 hours. How much money does she earn in 3 hours?

Directions: Solve using any method.

In your prize shelf at school there are two erasers for every 5 pencils. If there are 40 pencils, how many pencils and erasers are there combined?

The ratio of sunny days to cloudy days in the next 30 days is 7:3. How many **more** sunny days will there be?

Directions: Determine the value of the expression below.

$$\left(\frac{\cancel{2}^{8}}{\cancel{3}_{12}} - \frac{\cancel{1}^{3}}{\cancel{4}_{12}}\right) + \frac{4}{5} =$$

$$\frac{\cancel{5}^{25}}{\cancel{12}_{60}} + \frac{\cancel{4}^{48}}{\cancel{5}_{60}} = \frac{73}{60} = 1\frac{13}{60}$$

a. $\frac{3}{12}$

b. $\frac{5}{12}$

c. $\frac{8}{12}$

d. $1\frac{13}{60}$

Directions: Find the dimensions of the garden.

The total perimeter of a square garden is $\frac{96}{11}$ meters.

$$\frac{96}{11} \div \frac{4}{1} =$$

$$\frac{96}{11} \times \frac{1}{4} = \frac{96}{44} = 2\frac{8}{44} = 2\frac{2}{11}$$

Lucinda is knitting an outfit for her dog. She needs $\frac{3}{4}$ of a yard of fabric for his socks and $1\frac{1}{2}$ times more than that for his sweater. How much fabric does she need to make the outfit?

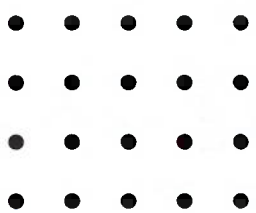
$\frac{3}{4} \times 1\frac{1}{2} =$

$\frac{3}{4} \times \frac{3}{2} = \frac{9}{8} = 1\frac{1}{8}$

$\frac{\cancel{2}^{6}}{\cancel{4}_{8}} + 1\frac{1}{8} = 1\frac{7}{8}$

Directions: Determine the value of the array by writing a multiplication sentence and then solving.

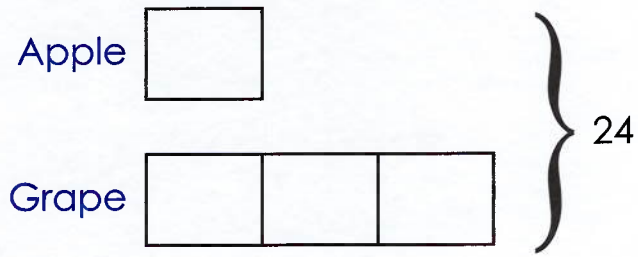
$\bullet = \frac{4}{9}$



$20 \times \frac{4}{9} = \frac{80}{9} = 8\frac{8}{9}$

Directions: Solve using any method.

If you need to make 24 cups total of your favorite party punch, how many cups of grape juice will you need?

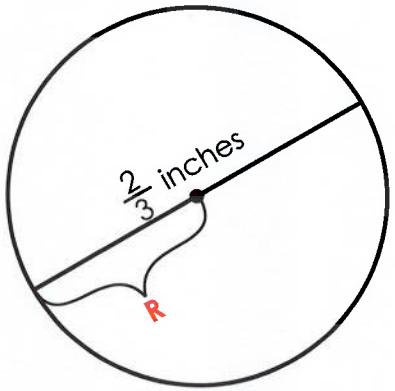


a. 6 cups
b. 18 cups
c. 3 cups
d. 9 cups

Sergio can type 1,340 words in 20 minutes. How many words can he type in 60 minutes?

Describe the steps used when multiplying fractions. Use the multiplication problem $\frac{3}{4} \times \frac{2}{6}$ in your explanation.

Directions: If the diameter is $\frac{2}{3}$, find the radius.

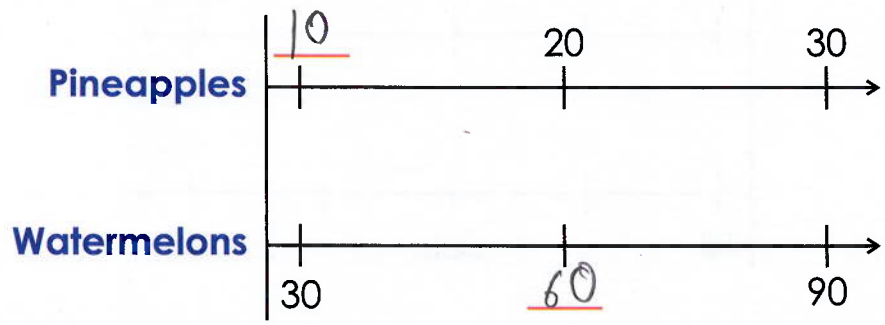


The radius of a circle is half of the diameter.

Explain when you would use a double number line. Give an example.

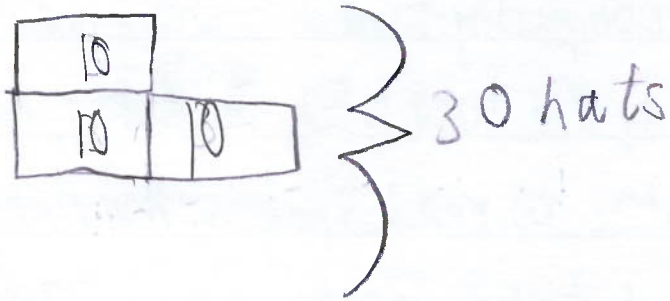
I would use a double number line for comparing two different units and knowing the total. Ex. It takes me two minutes to read 3 pages. How many minutes will it take me to read 27 pages?

Directions: Write the ratio represented by the double number line in lowest terms.



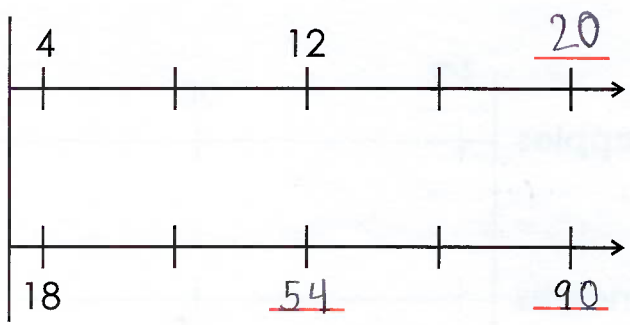
Directions: Solve using any method.

Julie and Arty have hats in a ratio of 1:2. Together they have a total of 30 hats. How many hats does each child have?



10 hats each

Directions: Complete the double number line.



Directions: Solve using any method.

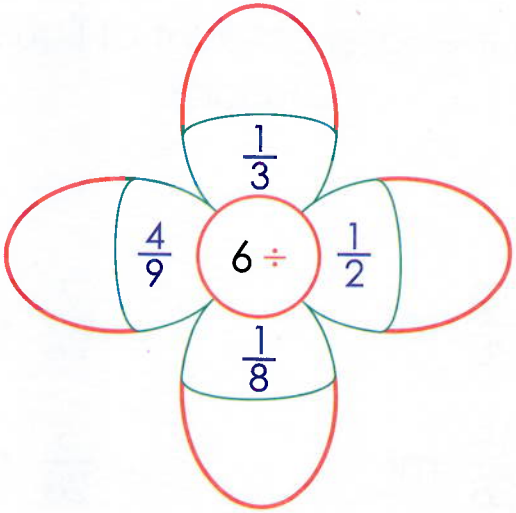
1. $\frac{3}{9} \times \frac{2}{4} =$

2. $\frac{1}{5} \times \frac{2}{6} =$

Marcos walked $3\frac{1}{2}$ miles. Dante walked $5\frac{3}{8}$ miles. How much farther did Dante walk?

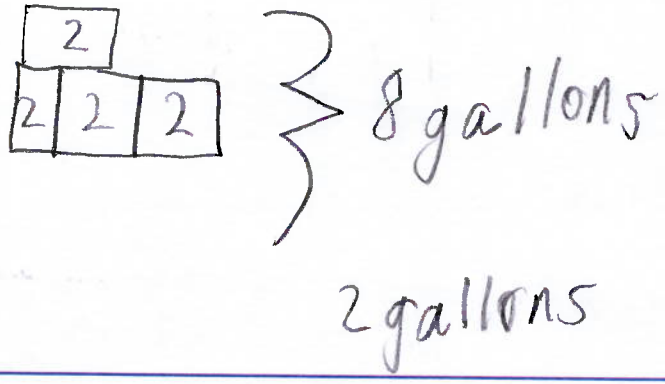
Jason walked $\frac{1}{5}$ of a mile from school to Carmen's house and $\frac{3}{8}$ of a mile from Carmen's house to his own house. How many miles did Jason walk in all?

Directions: Divide the number in the middle by each number on the petals.

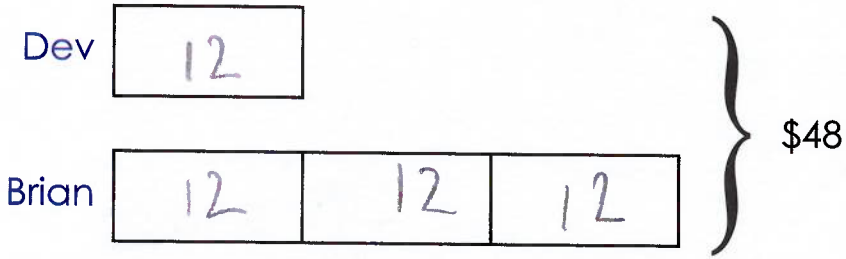


Directions: Solve using any method.

You need purple paint for your walls. You only have blue and red paint which you mix together at a ratio of 1:3 to make purple paint. How much blue and red paint do you use if you need a total of 8 gallons of purple paint?



Dev and Brian were paid $48 for dog walking at a ratio of 1:3.

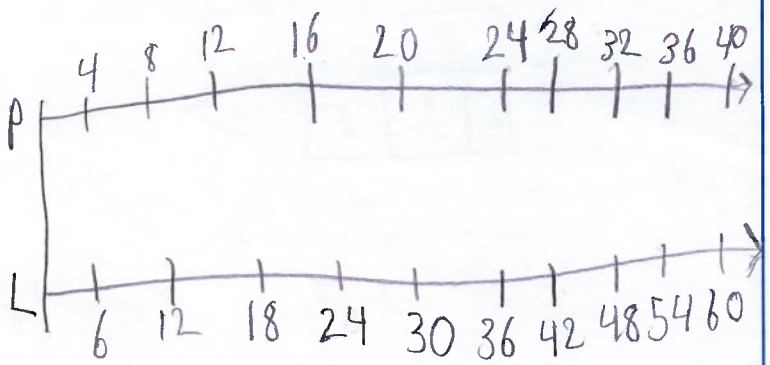


How much did each get paid per dog walk?

$12

Directions: Solve using any method.

In a video game you earn 4 points for passing 6 levels. If you passed 60 levels, how many points should you receive? Show your work by drawing a double number line.



(a) 18

(b) 27

(c) 40

(d) 48

40 points

GM5aW2.8

Level 5a

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Name

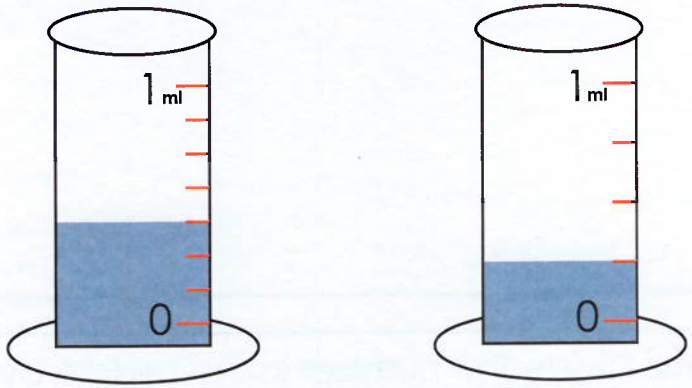
Date / /

Time ____ : ____ TO ____ : ____

Week 23 W6

Fractions: All Operations | **Ratios and Proportions**

In science class you need to combine the liquid from two different graduated cylinders.



Solution A Solution B

What is the total amount of liquid in the solution?

a. $\frac{12}{28}$ ml

b. $\frac{19}{28}$ ml

c. $\frac{7}{28}$ ml

d. $\frac{5}{28}$ ml

GM5aW6.1

Directions: Choose whether a double number or tape diagram would be best used to answer the question. Then explain your choice and solve.

An international call made from the US to Germany costs $3.50 per minute. If your mom made an 8 minute phone call, how much did it cost her?

☐ tape diagram ☐ double number line

Explain

_____

_____

_____

Answer: _____

Name

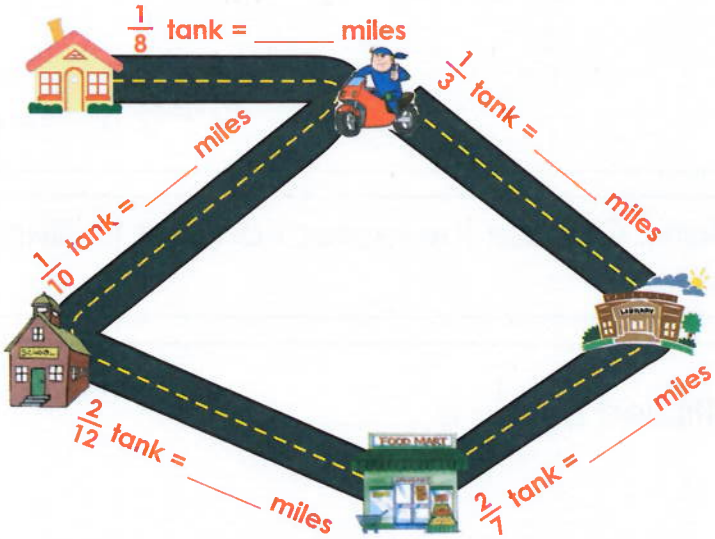
Date / /

Time ____ : ____ TO ____ : ____

Week 23 W3

Fractions: All Operations **Ratios and Proportions**

Leo's motorcycle has half a tank of gas. A full tank will take him 342 miles. Determine the amount of miles each distance is.



Can Leo go from his house to school to pick up his brother, and then to the grocery store for milk and back home on three-fourths a tank of gas?

A loaf of bread is 12 inches long. How many pieces can you cut to make $\frac{1}{4}$ inch pieces?

$12 \div \frac{1}{4} =$

$12 \times \frac{4}{1} = \frac{48}{1}$

48 pieces

Directions: Choose the correct answer to fill in the blanks.

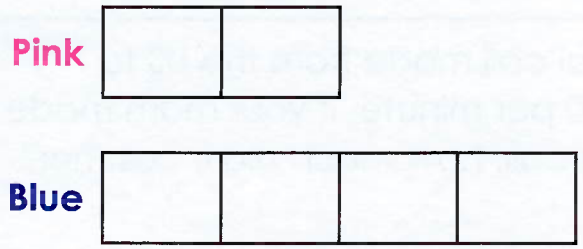
1. 9 multiplied by $\frac{1}{3}$ is _____

a. 2 b. 6 c. 3

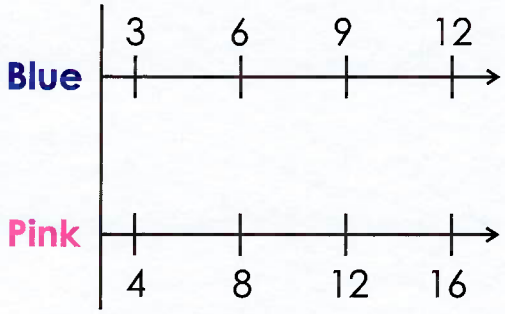
2. $\frac{1}{4}$ of _____ is 90.

a. 24 b. $\frac{1}{9}$ c. 360

Directions: Write the ratio of pink to blue represented in each image.



Ratio

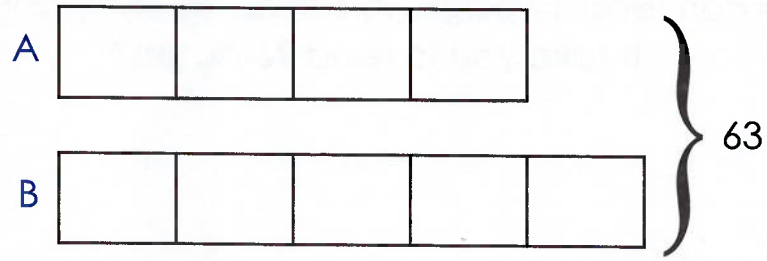


Ratio

Directions: Solve using any method.

A lemonade recipe calls for 4 tablespoons of sugar for every 2 cups of lemon water. If you want to make 20 cups of lemonade, how much sugar do you need?

Directions: Determine the value of each box.



Value A: ☐ Value B: ☐

Directions: Solve using any method.

Cale and Anthony each get a weekly allowance. The ratio of their allowance is 5:3. If the total allowance for both boys is $40, how much did each boy receive?

Directions: Solve using any method.

1. $5 \div \frac{3}{4} =$

$5 \times \frac{4}{3} = \frac{20}{3} = 6\frac{2}{3}$

$6\frac{2}{3}$

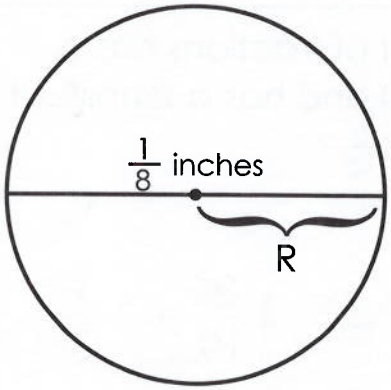
2. $3\frac{\cancel{4}\,12}{\cancel{5}\,15} + 2\frac{\cancel{1}\,5}{\cancel{3}\,15} =$

$\frac{57}{15} + \frac{35}{15} = \frac{92}{15} =$

$6\frac{2}{15}$

$6\frac{2}{15}$

Directions: The radius of a circle is half of the diameter.

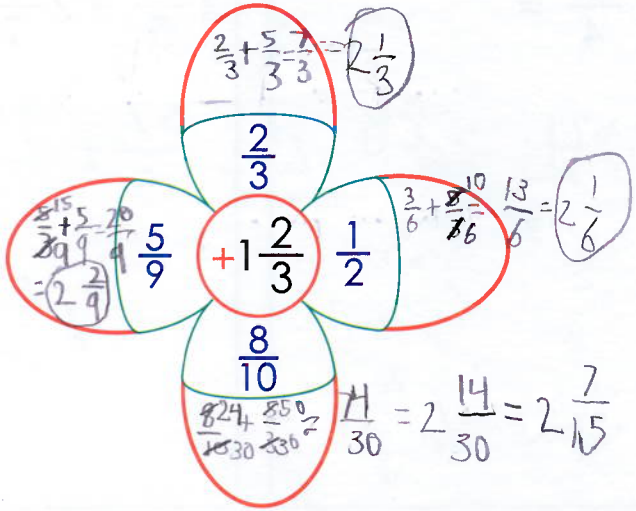


If the diameter is $\frac{1}{8}$ inches, find the radius.

$\frac{1}{2} \times \frac{1}{8} = \frac{1}{16}$

$\frac{1}{16}$

Directions: Add and write your answers in simplest terms.



Directions: Answer the question.

Which of the following set of fractions has a common denominator of 30 and has a simplified answer of $5\frac{8}{15}$?

a. $3\frac{4}{5} + 2\frac{3}{6}$

b. $6\frac{2}{7} - 4\frac{3}{5}$

c. $4\frac{2}{10} + 1\frac{1}{3}$

d. $7\frac{3}{13} + 1\frac{2}{5}$

Directions: Solve using any method.

You can read 12 pages in 6 minutes. How long will it take you to read 96 pages?

Dish soap is made at a ratio of 2:7 representing 2 parts concentrate for every 7 parts water. How many gallons of concentrate do you need to make 81 gallons of dish soap?

Your cat eats $\frac{2}{10}$ of a bag of cat food every day. How many days will 2 bags of cat food last?

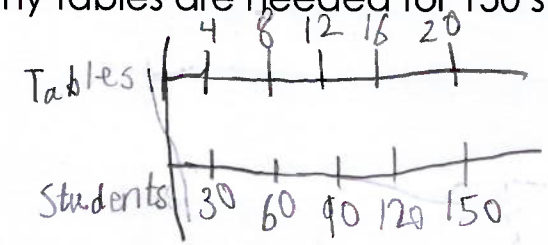
Directions: One problem has been solved correctly. Place an **X** over the wrong answer and solve.

a. $\frac{2}{4} + \frac{9}{16} = 1\frac{1}{16}$

b. $\frac{9}{13} + \frac{4}{26} = \frac{13}{11}$

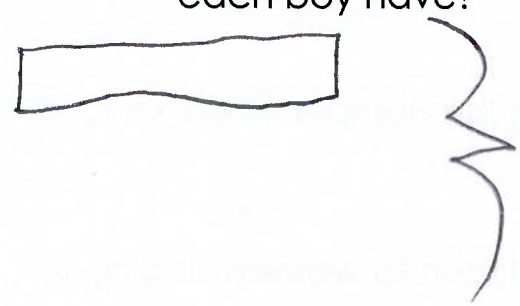
Directions: Solve using any method.

1. There are 30 students sitting at 4 tables. How many tables are needed for 150 students?



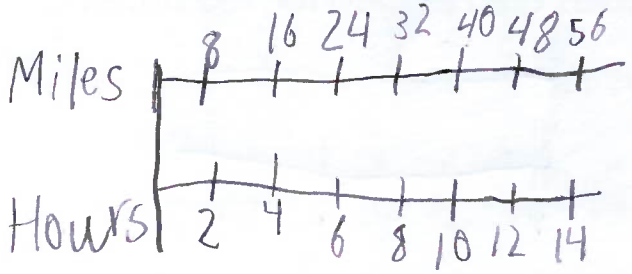
20 tables

2. Sam has 7 more stamps than Joe. They have 45 stamps altogether. How many stamps does each boy have?



Directions: Solve using any method.

Tony walks 8 miles in 2 hours. How long will it take him to walk 56 miles?



14 hours

Determine the scenario in which a tape diagram would be most useful.

(a) comparing the number of books to students in a class

(b) comparing men to women at a gym

(c) comparing cups of rice for pudding and cups of milk

GM5aW3.6

Directions: Solve using any method.

1

$2 \times \frac{2}{4} =$

2

$\frac{3}{8} \times \frac{2}{7} =$

Find the difference of $2\frac{1}{4}$ and $\frac{3}{6}$. Express in simplest form.

GM5aW5.3

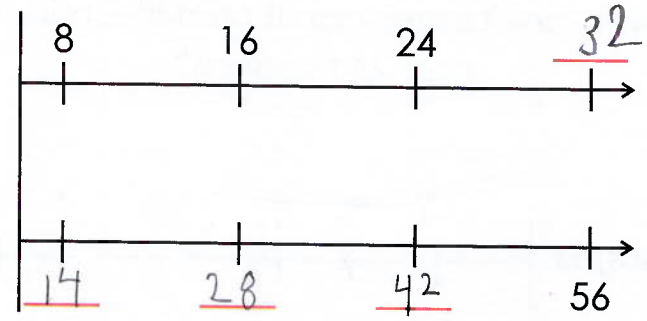
Directions: Convert each mixed fraction to an improper fraction. Then subtract the two fractions.

a. $4\frac{5}{8}$ =

b. $2\frac{3}{4}$ =

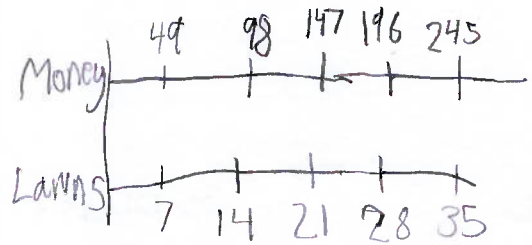
Maggie is $\frac{4}{7}$ as tall as Louie. Louie is $3\frac{1}{2}$ feet tall. How tall is Maggie?

Directions: Complete the double number line.



Directions: Solve using any method.

Steven earns $49 for mowing 7 lawns. How much will he earn for mowing 35 lawns?



$245

Level 5a

bestbrains LEARNING CENTERS

Name

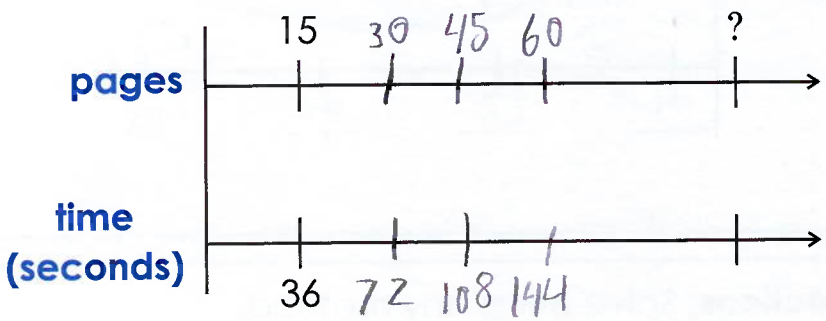
Date / /

Time ____ : ____ TO ____ : ____

Week 23 W5

Directions: Solve.

A photo copier can print 15 copies in 36 seconds. How many pages can it print in 2 minutes and 24 seconds?



a. 15 pages

b. 60 pages

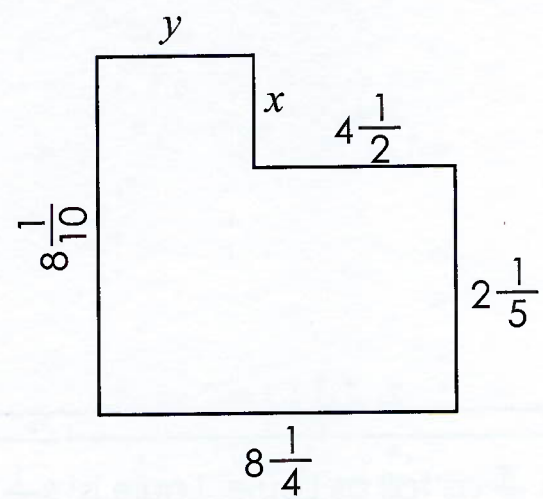
c. 36 pages

d. 144 pages

GM5aW3.8

Fractions: All Operations | **Ratios and Proportions**

Determine the values of x and y.



a. $x = 5\frac{9}{10}$ $y = 3\frac{3}{4}$

b. $x = 3\frac{3}{4}$ $y = 10\frac{3}{4}$

c. $x = 5\frac{3}{5}$ $y = 4\frac{1}{5}$

d. $x = 10\frac{3}{10}$ $y = 4\frac{1}{5}$

GM5aW5.1

Directions: Choose whether a double number or tape diagram would be best used to answer the question. Then explain your choice and solve.

An archaeologist found old dinosaur bones and teeth at a ratio of 4:1. If there were a total of 30 teeth and bones found, how many of each were found?

☐ tape diagram ☐ double number line

Explain

_____

_____

_____

Answer: _____

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 23 W4

Fractions: All Operations **Ratios and Proportions**

Directions: Reduce each fraction to lowest terms.

(1) $\frac{7}{35} =$

(3) $\frac{6}{50} =$

(2) $\frac{8}{16} =$

(4) $\frac{21}{30} =$

Directions: Determine if the fraction has been reduced to lowest terms.

$$\frac{10}{30} = \frac{5}{15}$$

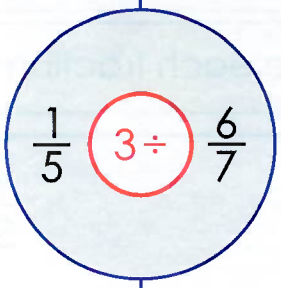
(1) What was the numerator and denominator divided by? _____

(2) Can the fraction be reduced further? If yes, continue to question 3 and 4. **Yes / No**

(3) What is the GCF of $\frac{5}{15}$? _____

(4) Write the new reduced fraction. _____

Directions: Divide the number in the middle by each number in the shaded area.



Directions: The table shows the length of three different strings.

String	String A	String B	String C
Length (in)	$1\frac{4}{7}$	$15\frac{1}{2}$	$10\frac{2}{7}$

1. What is the combined length of string A and B?

2. What is the combined length of string B and C?

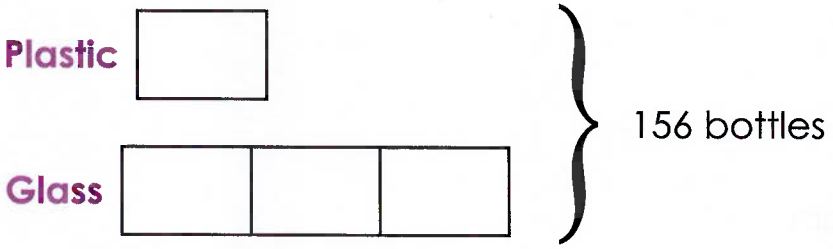
Directions: Use a tape diagram to answer the questions.

For a wall mural, 10 cans of white paint are needed for every 4 cans of blank paint. If 24 cans of black paint were used, how many cans of white were used?

A car can travel 96 miles on 8 gallons of gas. How far can the car travel on 16 gallons of gas?

Directions: Solve using any method.

Arnav and Saleem have 35 rolls of tape at a ratio of 3:4. How many rolls do they each have?



For science class Ekiko collected 156 bottles. There were 3 times as many glass bottles as plastic. How many glass and plastic bottles were there?

Directions: Solve using any method.

1. $3 \div \frac{7}{9} =$

2. $3\frac{1}{2} \times \frac{2}{4} =$

On average, a sloth moves $\frac{6}{25}$ kilometers a week. At this rate, how many kilometers will it move in one day?

For a science project you have $\frac{11}{12}$ of a yard of yarn. You cut off $\frac{2}{3}$ of a yard of the yarn. How much yarn do you have left?

a. $\frac{8}{12}$ of a yard

b. 1 yard

c. $1\frac{8}{12}$ of a yard

d. $\frac{1}{4}$ of a yard

Directions: Find the value of y.

1

$$8\frac{2}{7} + 3\frac{1}{8} = 11\frac{23}{y}$$

2

$$4\frac{y}{5} + 6\frac{9}{12} = 11\frac{11}{20}$$

Directions: Choose whether a double number or tape diagram would be best used to answer the question. Then explain your choice and solve.

Aunt Jemini's famous pancakes are 3 parts flour to 2 parts water. If she used a total of 18 parts flour, how much water does she need?

☐ tape diagram ☐ double number line

Explain

_____

_____

_____

Answer: _____